

CO3-AT1-Assignment

View Only

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks



Q2

Hospital Patient Registration –
User-Defined Exception

10 marks



Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks



Q4

Online Food Delivery – Multiple
Threads and Execution Order

10 marks



Q5

Manufacturing Plant –
Producer-Consumer Problem

Q2 Hospital Patient Registration – User-Defined Exception

10 marks

Develop a Java program for a hospital patient registration system. The program accepts the patient's name, age, and body temperature.

Create suitable user-defined exception classes for invalid age and invalid temperature. Validate the input before registration. Use try-catch-finally and display a registration completion message from finally.

Sample Test Cases:

Input / Scenario Expected Result Purpose

Arun, 35, 98.6 Patient registered successfully Normal case

Meena, 12, 98.4 InvalidAgeException Boundary / exception case

Ravi, 45, 105.2 InvalidTemperatureException Invalid input

Kumar, -2, 98.1 InvalidAgeException Invalid input

Your Code

```
1 import java.util.*;  
2 class InvalidAgeException extends Exception{  
3     InvalidAgeException(String message) {  
4         super(message);
```

Input

Arun
35
98.6

10 marks

5/5 answered

```

5      }
6      public static void main(String[] args){
7          Scanner sc=new Scanner(System.in);
8          try{
9              String name=sc.next();
10             int age=sc.nextInt();
11             double temperature=sc.nextDouble();
12             if(age<=0||age<13){
13                 throw new
14                     InvalidAgeException("InvalidAge:
15             }
16             if(temperature<95.0||temperature>105.0)
17                 throw new
18                     InvalidTemperatureException("In
19             }
20             System.out.println("Patient registered
21         }catch(InvalidAgeException e){
22             System.out.println(e.getMessage());
23         }catch(InvalidTemperatureException e){
24             System.out.println(e.getMessage());
25         }catch(InputMismatchException e){
26             System.out.println("Invalid input.");
27         }finally{
28             System.out.println("Registration comple
29         }
30         sc.close();
31     }
32 }
33 class InvalidTemperatureException extends Exception
34     InvalidTemperatureException(String message){
35         super(message);
36     }
37 }

```

Output

```

Patient registered
successfully
Registration completed.

```



✓ Submitted

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